

G D HRUTHIK

Boston, MA | +1 (508) 735-4181 | hruthik07022001@gmail.com | [LinkedIn](#)

AI/ML ENGINEER

PROFESSIONAL SUMMARY

AI/Machine Learning Engineer with 3+ years of experience in designing and deploying end-to-end ML pipelines, computer vision workflows, and scalable MLOps solutions. I am skilled in MLflow, Docker, Kubernetes, TensorFlow, and PyTorch, with expertise in moving models from research to production. Experienced across healthcare, biotech, and enterprise domains, including projects in clinical trial data pipelines, anomaly detection, fraud detection, and recommendation systems. Recognized for building AI tools that improved adoption by scientists and delivered measurable business impact through reliable, production-grade deployments.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, C++, R, SQL, Scala, Pandas, NumPy
- **Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Bagging, Boosting, Stacking), Hyperparameter Tuning (Optuna, Grid Search).
- **Deep Learning & Computer Vision:** CNN, RNN, LSTM, BERT, CodonBert, ProGen2 Transformers, Attention Mechanisms, Reinforcement Learning, CLIP, Image Classification, Object Detection, Fine-tuning Models, Diffusion Models
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, DeepSpeed, Flask, Django
- **Natural Language processing:** Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis, LangChain, RAG, Hugging Face Transformers, OpenAI API
- **Generative AI:** GPT-based LLMs, Diffusion Models, Prompt Engineering
- **Vector Search & Retrieval:** FAISS, Pinecone, Chroma, Weaviate
- **MLOps & Model Deployment:** MLflow, DVC, Docker, Kubernetes, Azure Kubernetes Service (AKS), Streamlit, Ray, TorchServe, FastAPI, Triton Inference Server, TensorFlow Serving
- **CI/CD & Workflow Orchestration:** Jenkins, GitHub Actions, Apache Airflow, Azure DevOps, Luigi
- **Cloud Platforms:** AWS SageMaker, GCP Vertex AI, Azure ML Studio
- **Visualization & Reporting:** Tableau, Matplotlib, Seaborn
- **Data Engineering:** Hadoop, PySpark, Azure Databricks, Data Factory, Synapse Analytics, Snowflake
- **Project & Collaboration Tools:** Jupyter, Google Colab, VS Code, Agile, A/B Testing, Requirement Gathering

EDUCATION

Master of Science in Data Science and Analytics

June 2025

New England College, Henniker, NH- GPA: 3.74

PROFESSIONAL EXPERIENCE

Sanofi, Boston, MA

April 2024 - Present

Machine Learning Engineer

- Worked with clinical teams to set up **ML pipelines** for genomic and clinical trial data, improving experiment reproducibility.
- Helped containerize models with **Docker and Kubernetes**, enabling scalable deployment across oncology and immunology studies.
- Implemented **model monitoring, versioning, and compliance checks** aligned with FDA and GxP requirements.
- Contributed to internal AI tools built with **Streamlit and FastAPI**, which increased adoption of AI solutions among scientists by **40%**.
- Integrated **EHR and biomarker data** into production pipelines, supporting translational medicine teams in biomarker discovery.
- Developed a **computer vision workflow** with CNNs and PyTorch for microscopy images, reducing protein misfolding detection time by **50%** compared to manual review.

Cognizant Technology Services, Coimbatore, TN, India

Aug 2021 - May 2023

Machine Learning Engineer

- Designed **real-time anomaly detection models** with Autoencoders and Isolation Forests, cutting equipment downtime by **20%** for industrial clients.
- Built **predictive maintenance pipelines** using Random Forest, Kafka, and Azure Stream Analytics to forecast equipment failures.
- Deployed a **fraud detection solution** in Azure ML that achieved inference speeds under **120ms** and improved detection accuracy.
- Supported an **e-commerce recommendation system** in TensorFlow, leading to a **15% increase in click-through rates** during A/B testing.
- Collaborated on **end-to-end ML operations pipelines**, standardizing containerization and monitoring across projects.